Smart Control Tower · Architecture & Workflow Walkthrough

Amber Press Shop · Evidence-Linked Operational Review & Deterministic AI Intelligence

AXIOM: APP CALCULATES, AI EXPLAINS | EDGE-BRIDGE ARCHITECTURE | CASE STUDY & TECHNICAL BRIEF

Operational Challenge

Plant floor monitors display near-live DFOS telemetry, but daily leadership reviews suffered from manual Excel consolidation across 3 separate workbooks (Production, Downtime, Rejection). Result: unverified OEE metrics (>100%), missing loss reasons, and zero action follow-through.

Engineering Solution

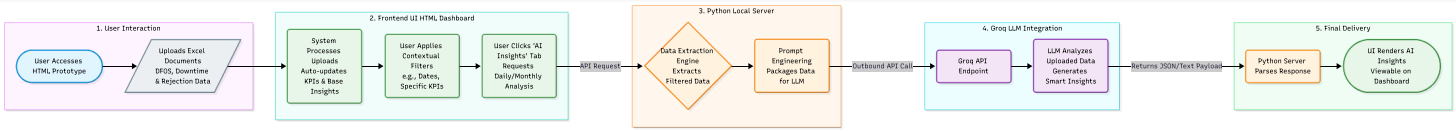
A decoupled, zero-installation Smart Control Tower that unifies multi-report schemas client-side, computes deterministic KPIs & Top 10 Paretos in JavaScript, and routes bounded statistical summaries through a tokenized Python bridge to an LLM fallback chain.

Business & Quality Impact

Review preparation dropped from 90 mins to under 10 seconds. Eliminates AI arithmetic hallucination entirely, protects factory privacy (zero raw rows leave browser), and automatically establishes mathematical reconciliation before managerial sign-off.

LOGICAL ARCHITECTURE & END-TO-END WORKFLOW PIPELINE

Source: Christmas Shopping Decision-2026-10-12-073041.svg (Mermaid Architecture Pipeline)



STAGE 1
User Interaction

- **Multi-File Ingestion:** Drag-and-drop Production, Downtime & Rejection XLSX/CSVs.
- **Preflight Scoping:** Schema check, date range & machine alias normalization.
- **Context Filters:** Interactive shift, machine cell, model & date slicing.
- **AI Trigger:** On-demand Daily / Monthly briefing dispatch.

STAGE 2
Frontend Dashboard

- **Client Parsing:** In-memory SheetJS parsing (zero server upload).
- **Deterministic KPIs:** Auto-computes Attainment, Availability, Scrap %.
- **Top 10 Paretos:** Ranks loss minutes & reject pieces deterministically.
- **Prompt Packaging:** Minimizes context; excludes sensitive employee remarks.

STAGE 3
Python Local Bridge

- **Local Endpoint:** Pure Python stdlib server binding strictly to `127.0.0.1:4173`.
- **CSRF Token Gate:** 32-byte cryptographically random session bearer token.
- **Payload Throttling:** Enforces <120KB payload limits for cost/latency.
- **Response Parsing:** Normalizes vendor JSON streams into Markdown.

STAGE 4
LLM Resilience Layer

- **Fallback Routing:** Primary OpenAI → Mistral → Groq Edge (`gpt-oss-120b`).
- **Custom User-Agent:** App-specific signature bypassing edge rate-limit walls.
- **Strict Guardrails:** Explains facts; forbidden from recalculating numbers.
- **Zero Retention:** Disables server training with `store: false`.

STAGE 5
Final Delivery

- **DOM Sanitizer:** Renders responsive cards, clean tables & badges.
- **Side-by-Side Proof:** Displays verified Top 10 tables alongside AI narrative.
- **Math Guardrail:** Strips downtime % if Total ≠ Planned + Unplanned.
- **Action Tracking:** RCA case creation with owners & effectiveness checks.

Core Engineering Safeguards

Axiom: App Calculates • AI Explains

1. Mathematical Shield

KPIs, percentages, and Top 10 rankings are calculated in JavaScript. The LLM is strictly prohibited from doing arithmetic, eliminating hallucinations.

2. Air-Gapped Privacy

Raw workbooks and worker remarks stay in browser memory. Only aggregated statistical summaries are sent to the LLM API; zero factory PII escapes.

3. Multi-Provider Circuit Breaker

If OpenAI hits quota exhaustion or HTTP 429, the bridge silently cascades to Mistral, then Groq Edge, ensuring 99.9% uptime for morning plant reviews.

Production Evolution Roadmap

PHASE 1 • LIVE PILOT
Local-First MVP
Client-side XLSX parsing, stdlib server bridge, session storage, and manual Excel drop.

PHASE 2 • NEAR TERM
Hosted Team App
FastAPI + PostgreSQL, Entra ID SSO, immutable batch lineage & durable action register.

PHASE 3 • ENTERPRISE
Automated DFOS Sync
Direct read-only DB view/API, 3-minute IoT micro-sync, and automated anomaly alerts.